Sonic Agency: A Group Autoethnography of Technology-mediated Performance Practice by Deaf and Hard of Hearing Musicians

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Abstract

While many assistive devices and accessible technologies aim to increase access and participation of d/Deaf and Hard of Hearing (DHH) audiences, DHH people's active participation in music performance remains limited. Through a 15-month-long group autoethnography, we present a case study of a Deaf-owned music and culture institution in a mixed-hearing, technology-mediated performance practice. We unpack how technologies are adapted and developed by DHH musicians and highlight the existing and emerging challenges of mixed hearing creative spaces. We use retrospective accounts, field notes, and performance artifacts to reveal the methods we developed for more inclusive music co-creativity. We lastly present three essential strategies that support DHH individuals' sonic agency and access, with or without prior music knowledge, applicable by different actors who partake in the DHH or mixed hearing music performance ecosystem.

CCS Concepts

• **Human-centered computing** → **Accessibility technologies**; Accessibility theory, concepts and paradigms; • **Applied computing** → *Performing arts*.

Keywords

musical accessibility, autoethnography, creative practice, deaf studies, performance-led, active participation

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1 Introduction

Many assistive devices and accessible technologies aim to increase access and participation of d/Deaf and Hard of Hearing (DHH) audiences in music performance as *listeners*. However, research remains limited when promoting access for DHH people's active participation as artists, designers, and developers [17, 54]. Specifically, when we consider the actors in the design and technology development process, DHH people with musicianship skills are little represented [31, 32].

Recent research has focused on increasing active participation of disabled musicians in design through participatory [4, 21], bespoke [46], and dialogical [77] approaches. Despite the efforts toward a more inclusive design practice, DHH individuals partook in the design process with limited agency, and their creative role remained relatively passive. Few research studies collaborated with DHH individuals as creative actors. Ohshiro and Cartwright [58] conducted survey studies to identify the trends in technology use by DHH audio engineers. Cavdir [11] carried co-design session for music-dance collaboration with a Deaf performer. Other researchers explored DHH individuals' relationship with music by centering them in the studies through the lens of song signing [17, 76] or sensory substitution [22, 36]. Outside Human-Computer Interaction (HCI), some researchers sought to understand how DHH musicianship is learned, developed, and practiced [6, 18, 27].

Despite the shift in accessibility research from *designing for people* to *designing by people* [3, 9, 43], deaf-led research remains rare at the intersection of HCI and musical accessibility. Music access for DHH individuals remains restricted within passive audience roles, rather than exploring active musicianship.

In this study, we examine the experience of DHH musicians, technology developers, and designers who advocate for their sonic agency and their communities' rights to an equitable musical experience. Through a 15-month-long group autoethnography, we present a case study of a Deaf-owned music and culture institution in a mixed-hearing, technology-mediated performance practice. Utilizing retrospective accounts, field notes, and performance artifacts, we draw perspectives on how technologies are integrated into creative musical practice through real-life performance scenarios.

Our contribution from this research is three-fold. First, we unpack how technologies are adapted and developed by DHH musicians. We provide a summary of our current practices with custom design and commercially available assistive musical technologies through three main integrated sensory modalities: haptic displays, audio-reactive light and visuals, and gestural interfaces. Second, we reflect on how DHH musicians define sonic agency in hearing-centric communities and articulate three underlying practices: our role as music technology developers, hacking and re-purposing practice as a form of resilience, and performance as a tool to gain visibility, access, and sonic agency. Third, we highlight three main challenges of misalignment of musical expectations, unfamiliarity with musical concepts, and technological limitations that existing technologies impose. We detail such limitations that derive from the lack of Deaf technologists' participation in design and development. To cope with these limitations, we provide our strategies of prioritizing music composed by Deaf musicians, adapting a more somatic approach to novice music practice, integrating graphical and gestural notation systems, and engaging in bespoke design and appropriating technology. These strategies were devised throughout this practice-based research to cope with some of the challenges that Deaf musicians and audience members experience in mixed hearing co-creative spaces.

2 Related Work

In this section, we first review foundational research on musical accessibility across multiple HCI literature as a background for our work. We also discuss specific background work on accessibility research for music and audio technologies designed specifically for, with, or by Deaf and Hard of Hearing individuals. Lastly, we survey the method of autoethnography from its adaptation in HCI to its applications in accessibility research.

2.1 Music Accessibility in HCI

Music accessibility research extends across multiple research communities, from Human-Computer Interaction (HCI) to New Interfaces for Musical Expression (NIME), and across various disabled populations, such as Blind and Low Vision (BLV), neurodivergent individuals, individuals with mobility restrictions, and individuals with cognitive disabilities. In this section, we outline the accessibility research for musical expression, learning, and participation for a broader group of users. In the following section, we survey musical accessibility more specifically from the perspective of Deaf and Hard of Hearing individuals and communities. Both HCI and music researchers closely studied musical interaction for visually impaired individuals. Ahmed et al. initially designed a music notation system for BLV musicians and composers, SoundCells, that simultaneously outputs visual and braille notation while providing audio feedback [2]. Payne et al. further conducted a 6-week study using SoundCells, investigating how vision ability, along with musical experience and use of assistive technologies, impacted musical access, creativity, and expressivity [60]. While Ahmed adapted braille notation systems for music composition [2], Manenti and Dal Ri conducted workshops with BLV musicians to design graphical tactile scores as an alternative to the Braille music notation [51]. Payne et al. carried assistive music technologies to the music classroom to improve the accessibility of live coding ensembles to both BVI and high school learners [61]. More specifically for music training and practice, Lu et al. investigated the multimodality of assistive musical technologies for music reading and memorization [45].

Designing accessible musical interactions for mobility restrictions and physical limitations is among the most commonly studied topics in accessibility research [26]. Sporka et al. also worked with three music composers and producers who have motor disabilities to identify the access opportunities and limitations of existing assistive music technologies [71]. Similar to Sporka, McMillian and Morreale explored mobility restrictions by integrating different occupational therapy methods into instrument design considerations from sensing technologies to movement and functional limitations [53]. McCloskey et al. studied accessible button/switch controllers for digital musicians with cerebral palsy to explore more inclusive gestural control [52]. Similarly, Payne et al. integrated eye tracking into accessible musical instrument design for people who experience severe physical limitations and rely on eye trackers for computer access [62].

Much research also focused on increasing musical accessibility for individuals with neurodivergencies [26]. Jarvis Holland prototyped an ensemble of adapted instruments for artists with intellectual or developmental disabilities [40]. Lobo et al. developed a mobile system to support therapists "for teaching and tracking socio-communicative behaviors in children with neurodevelopmental disorders during music therapy sessions" [44]. Similarly, Forster et al. addressed children with complex or intellectual disabilities in special education needs classroom settings [29]. They designed a block-based musical interface to include children in musical activity without prior music knowledge required, providing visual feedback and passive haptic feedback from the materiality of the interface without external actuation.

Similar to the diversity of disabilities that the musical interfaces addressed, researchers also adapted different methods with varying levels of disabled individuals' engagement and participation, from bespoke design [46] to more participatory methods [4, 11, 21] to dialogical methods [77]. Although researchers advocate for participatory methods or centering the disabled person's lived experiences [70], the disabled individuals' participation in the design process and the researchers' interpretation and adaptation of the methods vary significantly [77].

On one hand, more user/musician-centric methods have been increasingly adapted [69]. On the other hand, accessibility research and designs conducted by disabled researchers and musicians are significantly limited. The literature includes a single autoethnography of a disabled musician with physical limitations [53] and could benefit from more research based on disabled musicians' first-person experiences of design, creative practice, and performance. Deaf experiences are even more reduced at the intersection of HCI, music, and accessibility research. In the next section, we detail the research for musical accessibility, specifically for Deaf and Hard of Hearing (DHH).

2.2 Music Accessibility and DHH

In Frid's survey on accessible musical instruments from 2019, no studies addressed deafness, and only %6 of all studies included worked with musicians with hearing impairments [26]. Although this statistic has changed in the past five years [11, 12, 17, 36], research on Deaf musicianship still remains limited. Research rather focuses on DHH individuals as users or music consumers rather than creators. For example, Choi et al. integrate customizable cross-modality into exploring musical concepts for DHH music listeners' appreciation [16].

To support DHH engagement in music, much research explores assistive modalities such as visual [14, 22, 33, 65] or vibrotactile representations [67]. These representations often provide musical information to substitute for the missing senses, while some utilize these stimuli to enhance this user group's musical experience. Nanayakkara, Wyse, Ong, and Taylor develop assistive devices with visual and haptic displays [56]. Similarly, Fourney and Fels provide visual feedback to inform "deaf, deafened, and hard of hearing music consumers" about musical emotions conveyed in the performance, extending their experience of accessing the music of the larger hearing culture [25]. Burn applies sensory substitution in designing new interfaces using haptic and visual feedback for "deaf musicians who wish to play virtual instruments and expand their range of live performance opportunities." [8]. He highlights the importance of multisensory integration for Deaf musicians when playing electronic instruments and integrates additional feedback modalities into virtual instruments. Similarly, Richards et al. [68] develop a wearable, multimodal harness to study tactile listening experiences with hearing participants. These experiences are delivered via extratympanic sound conduction and vibrotactile stimulation of the skin on the upper body.

Iijima et al. explored different gestural interactions on the smartphone for augmenting digital musical instruments while providing tactile feedback to DHH users [36]. Cavdir and Wang [13] explore new compositional and performance paradigms to provide a more physically felt musical experience through movement and haptic modalities. Cavdir [11] presents a co-design study with a Deaf dancer to develop a multimodal wearable haptic interface and a cross-modal vocabulary for a dance-music collaboration. In a survey study, Ohshiro and Cartwright collected DHH audio producers' experience [58]; however, their participants took a more speculative approach to technology support than reflecting their ongoing practice with assistive technologies. The DHH music technologies that engage in DIY practices for new musical interface design remain underrepresented in the HCI and music accessibility research. Similarly, only a few research studies explore the music created, listened to, and appreciated by Deaf musicians and audience across genres such as Deaf hiphop [6] or practices such as Deaf Rave [41, 59]. In this paper, we highlight DHH music creators and technologists' motivations, perspectives, and experiences as we identify them within Deaf Jazz thanks to jazz practice's openness to improvisation and experimentation.

2.3 First-person Methods

First-person account originates from experience on the body, or the *lived experiences* [55, 63, 64, 73], focusing on our perception, understanding, and experience of the outside world and differs from a third-person perspective of seeing the body as an external object in the world [35]. Similarly, autobiographical or biographical design methods can reveal participants' first-person perspectives in design research [5, 34]. Inspired by the following background literature, we ground our work on group autoethnography and autobiographical design within first-person approaches.

2.3.1 Autoethnography. Autoethnography, as a qualitative, first-person method, describes and analyzes personal experiences of a specific cultural context or phenomena [24, 66]. It encapsulates a holistic representation of the autoethnographer, from personal experiences to their cultural identity and personal backgrounds [20], investigating the studied phenomenon in its cultural context. By using methodological tools that are drawn from qualitative, ethnographic, narrative, and arts-based research [19], an autoethnographer connects their individual experiences to a wider cultural context. Among other first-person methods [23, 30, 49], HCI researchers have been integrating autoethnography to acknowledge the researcher's role and to explore different forms of observing and describing personal experiences with technology [48].

As accessibility researchers and disability studies scholars, Hofmann et al. provided their first-person experiences through a reflexive analysis surrounding misunderstandings of their own disabilities and oversimplification of disabilities. Similarly, Janicki et al. [39] articulate their lived experience with chronic illness within the context of more-than-human design and disability theories, expanding more-than-human design practices through disability perspectives.

In accessibility research, autoethnography has been increasingly adopted as a method to report the researcher's first-person experiences. Jain et al. [37] were among the first researchers to unpack the experience of traveling solo as a disabled individual. Similarly, Zeidieh [78] and Stephens et al. [72] focus on the experience of traveling as a disabled person from an autoethnographical account and discuss how technology can provide independence to the visually impaired individual as well as lack of systematic support and assistance for people with visual impairments [10]. Similar to travel, relocation experiences pose significant challenges for individuals with accessibility needs [75]. Yildiz et al. describe the first author's personal experience as a researcher with a mobility disability to understand present and new access needs, interpersonal accessibility, and digital access and connections with local disability communities to rethink "accessibility on the move." The authors further provided design implications to support the relation and travel of disabled individuals, extending to academic communities.

Researchers, who track their experience "on the move," from traveling and transportation to relocation experiences, reveal the challenges and opportunities in providing assistive systems in public spaces, various sociocultural contexts, and outside their comfort zones. From a different perspective on interdependence and assistive technology, Fusenegger and Spiel focus on personal use of assistive devices, providing insights on the first author's lived experience around the necessary use of assistive technologies (AT) on a daily basis, revealing the socio-technical dependencies on AT [28]. Fusenegger tracked their daily use of multiple assistive technologies to highlight how much AT supports the researcher's everyday, social, professional life, and well-being. In a similar study, Wu et al. apply autoethnography to investigate the experience of a person who stutters (PWS) when using videoconferencing technologies [74].

The duration of autoethnography, often as a longitudinal study, varies significantly based on the researched phenomena, methodological adaptations, and researchers. While Yildiz et al. engaged in the autoethnographic data collection for three months, some researchers collected data from individual instances or periods over a longer study time. Cassidy et al. collected data from 13 instances over a 2.5-year-long period [10], and Jain et al. [37] focused on the two distinct travel periods from the first author's life. Most longitudinal autoethnographies captured isolated instances rather than engaging in everyday data collection from personal experiences throughout the process. For example, Wu et al. captured the researcher's experience qualitatively and quantitatively after each video conference meeting over a 15-month-long autoethnography [74]. A shorter period, just as in Yildiz et al.'s 3-month-long [75] or Fussenegger's and Spiel's 10-day-long studies [28], allows for continuous data collection and reflection. Although data collection focuses on individual periods in the former cases, autoethnography reflectively continues after such personal experiences, and researchers continuously engage in reflexive writing and retrospective writing.

2.3.2 Group Autoethnography. Although Wu et al. they provide "a unique and complementary perspective beyond more traditional user-centered design and research methodologies," they recognize the challenges of the method since autoethnography can present an inherent subjectivity and bias, difficulty of balancing the personal and the analytical voice, an emotional intensity throughout the process placing the researcher in a vulnerable position [74]. To address this predominant meta-narrative and subjectivity rooted in deeply personal perspectives, researchers proposed involving other researchers to approach the shared phenomenon from multiple perspectives [57]. Jain et al. [38] adapt Norris et al.'s approach by conducting a trio-ethnography to retrospectively understand their experiences in academia, specifically the graduate school experiences as people with disabilities.

In their duoethnography, Abromonovich and Patitsas collected 12 journal entries each, and between each journaling session, they aligned their experiences by commenting and responding on each other's journals for clarification [1]. Similarly, Jain et al. commented on each other's retrospective account and field notes in a shared document, "relating to the experience, challenging the assumptions, offering revisions and alternate viewpoints" [38].

Compared to duo- and trio-ethnographies, group autoethnographies, which are larger than three autoethnographers, follow similar approaches since group autoethnographies provide opportunities to reflect on group thinking and experience collectively through more systematic ways of aligning experiences and reflections and challenging assumptions and misconceptions. Chen et al. employ group autoethnography of a hearing individual's experience in mixed-hearing ability higher education settings [15]. The authors emphasized that the "group" element in group or collective autoethnography becomes more pronounced in the data collection phase since each researcher collects their "experiences individually yet within the context of the collective experience." They also report that they gathered for in-depth reflection meetings when analyzing the data, while the collaborative spirit emerged more evidently in the "meaning-making phase", ensuring that the group members' insights collectively shape the interpretation and conclusions of our writing [15]. Similarly, Mack et al. applied this method to a larger group autoethnography setting with eleven authors to understand the experiences in the mixed ability and diverse experience virtual environments, combining fieldnotes and retrospective accounts [50].

3 Methods

We provide background on group autoethnography, the team composition and our positionality, and our data collection and analysis methods.

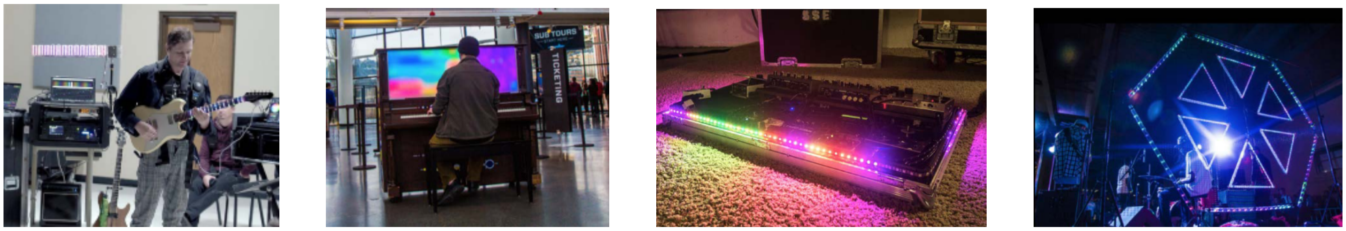
3.1 Team Composition and Posionality

CymaSpace [1] is a Deaf-owned technology hub whose mission is to increase media, arts, and culture accessibility for Deaf and Hard of Hearing (DHH) communities. Through five programs, they support Deaf communities' access to arts and culture: Research, education, outreach, entrepreneurship, and practice. Because diversity in hearing, experience of living with assistive devices, and exposure to music have close connections to individuals' participation and appreciation of music, in this section, we provide the musical, technical, and hearing backgrounds of the researchers [2], including their cultural associations and communication languages, to contextualize our findings in relation to the mixed ability performance ecosystem.

DC is an accessibility researcher and a musical instrument designer, studying the integration of sign languages into design space and designing new accessible interfaces for dance and music performance. She culturally associates with the hearing communities and actively collaborates with DHH artists. As a music and dance technology developer, she contributes to the design and research of CymaSpace's universal music design team as the team's academic mentor. She has a music and engineering background, performs with custom-designed musical interfaces, and composes with musical haptics. She is currently an American Sign Language (ASL) student and communicates with spoken and written English.

DS is a deaf technologist and lead design engineer who had profound hearing loss from birth. He associates with the lowercase d, deaf culture, due to being raised in a hearing family and cultural disagreements with the uppercase D, deaf communities. He only developed an interest in music and performance in 2023, starting with his collaboration with the CymaSpace organization. As a first-generation student, he is currently working towards his associate degree in Music and Sonic Arts. He performs percussion and custom sonic interfaces with hearing and Deaf musicians and develops his custom-designed assistive music technologies. He communicates with ASL and written English.

[1] https://www.cymaspace.org/
[2] Initials are used to refer to the co-authors.



(a) (b) (c) (d)

Figure 1: Sound reactive LED-based visualization: (a) a performer plays with the visualization system embedded on the guitar fretboard, (b) pitch and amplitude content of music are mapped to light, color, and movement during piano performance, (c) sound reactive LED strip is assembled into a sound board for visual feedback on sound effects, (d) sound reactive LED structure displayed on stage during concert performance. Image credits for (a) belong to Eric Buchner and for (b) through (d) to CymaSpace.

MB , the executive director of CymaSpace, is a Deaf/Blind musician. He uses hearing aids for a small band of audio perception due to progressing hearing loss, which started at birth and developed in his teenage years. Myles associates both with Deaf and hearing cultures. Growing up in a musical family, he has more than 20 years of music experience, playing the electric guitar, performing and touring with bands, and composing. With his background in UX/UI design, he collaborates with the team's designers on custom assistive music technologies, specifically for music performance and sensory substitution. He communicates with ASL, spoken, and written English.

ST is a hearing music producer, audio engineer, computer scientist, and researcher with autism. He started experiencing hearing damage in his teenage years, such as hyperacusis and tinnitus. He associates with the hearing culture and closely collaborates with DHH musicians and community members. Thanks to his experience, he manages the music studio and multimodal sensory setup during practice sessions and contributes to developing music educational materials for Deaf students. He also performs the percussion instruments and electronics. He communicates with ASL, spoken, and written English.

NH is a Deaf Musician and project lead of the universal music design team at CymaSpace. Growing up with hearing aids since the age of 3, NH progressively lost his hearing at 16. Five years after losing his hearing, he received a cochlear implant (CI) in one ear. He wears his implants when listening and speaking; however, during Deaf community events, he prefers to keep them deactivated for cultural reasons. He associates both with Deaf and hearing cultures. As a child, he actively participated in music practice and performance by playing various musical instruments, including the piano and trumpet, and participating in the marching band, as well as singing in touring music performance groups. He currently performs as a DJ and composes music using synthesizers and electronics. He communicates with ASL, spoken, and written English.

The team members have different cultural associations regardless of their hearing abilities. They collaborate on designing new assistive music technologies, composing, creating educational materials, and performing music. During the study, they communicated through ASL, sign interpretations, written, and spoken English. In virtual meetings, they only used video conferencing applications that allowed for accurate live captioning.

3.2 Background on the Performance Ecosystem

Four members of the team (DS, MB, NH, and ST) formed a Deaf-led ensemble, practicing technology-integrated music learning, co-performing, and composing, in two settings: practice sessions in a music recording studio and on tour across three local educational institutions.

3.2.1 Studio Sessions. The group gathered periodically in a music studio to co-perform as a small ensemble. ST led the studio sessions by mastering and engineering audio signals to multimodal channels, structuring the studio setup for increased collaboration and technology integration, and preparing educational and compositional materials in the form of etudes. These educational and compositional materials were designed specifically for Deaf musicians through the practice sessions and rehearsals for public events in the studio. He also recorded the Deaf musicians' compositions and participated as a percussionist in the performance sessions. Both DS and MB designed custom interfaces for musical control, expression, and communication between co-performers that are integrated into the studio setup (as detailed in Section 4). NH and MB created audio-haptic compositions for the ensemble's co-performance sessions. DC participated in facilitating the group discussions and organized group reflections on the individual performance sessions. She also collaborated in designing the custom gestural musical interfaces that were integrated into the studio's assistive technology setup. All Deaf musicians participated in practicing new musical etudes, co-performing, and improvising in a mixed-hearing performance ecosystem.

3.2.2 Tour and Outreach. The ensemble participated in a performing tour with two additional hearing musicians in December 2024, visiting three local institutions for four days of performance and outreach activities. During the tour, they performed both as an ensemble and in collaboration with the local jazz and voice orchestras for mixed-hearing audiences. Each performance was complemented with workshops and community events, providing participants with an opportunity to engage hands-on with haptic technology, experience audio-reactive visuals, and explore new interfaces for musical expression. The workshops were tailored to the music students and DHH community.

3.3 Data Collection

Data collection began in January 2024 and continued until March 2025. Each member collected their experiences through (1) retrospective account [24, 47], (2) fieldnotes [66] collected after studio and performance sessions, and (3) other artifacts, including memos, audio and video recordings, and personal reflections. The group aspect of this autoethnography was more pronounced in this stage since each member documented their experiences individually within a shared music studio environment. Some of the fieldnotes and personal reflections include the narrative of performances or community engagement events, such as specific tour days, outreach events in local schools and Deaf communities, or specific studio sessions. The retrospective accounts were collected based on the autoethnographer's experience with music learning, performing with technology, performing with others, or developing technology.

We initially developed a retrospective account of experience with music and technology development before more structurally collecting data in January 2024. These accounts included digital notes and correspondences of experiences in music learning, performance, technology use, community associations and outreach, and engaging in assistive and accessible technology design. To support recollecting these experiences, we referred to email correspondence, design logs, and performance recordings. These past experiences were also shared and discussed in group meetings, enabling other members to relate and reflect on each other's experiences. Fieldnotes were collected after studio sessions and after performances during the tour. In addition to collecting personal reflections, the authors shared their experiences, self-observations, and reflections in a shared form. They articulated specific musical activities from the sessions—including different musical instruments, assistive and custom technologies—performance dynamics, and expectations and takeaways. They also articulated their somatic state, including physical, emotional, and mental, before and after the studio sessions.

In addition to keeping fieldnotes as a musician, as the studio manager, ST documented his observations of the performance dynamics between co-performers and musical content as the studio manager. The collected data was reviewed one-on-one between DC and ensemble members and, again, collectively in group sessions. At the same time, she collected her self-reflexive account on the design, compositional, and performance experiences in addition to reflection on her correspondences with the ensemble. Although DC organized follow-up and journaling structures, the frequency of data collection occasionally fluctuated due to the nature of the studio practice (e.g., lasting until late at night) and to prevent cognitive overload related to varying access needs of the ensemble. Similarly, ST's data collection was interrupted for a month in December 2024 due to a medical leave.

3.4 Data Analysis

Following the approach adopted by Chen et al. [15] and Mack et al. [50], DC collected and organized the data from the retrospective accounts, fieldnotes, and reflections on the audio-visual artifacts. The group discussions were transcribed and included in the analysis. The collected data was analyzed through an inductive thematic analysis similar to Braun and Clark's approach to a more flexible and recursive process of codifying data and defining themes [7]. Starting the analysis, the authors initially created six high-level, fourteen low-level axial codes, which increased to eight high-level and fifteen low-level axial codes after collectively revising the codes. To prevent misinterpretations, these codes were reviewed with all authors in group meetings to align interpretations and to include additional reflections. These group sessions allowed for critically revising the codes by adding, removing, or merging them. Finally, the codes were combined into four overarching themes presented in Section 5: music technologies in practice, sonic agency and accessibility, challenges and access barriers, and strategies for increased inclusion in mixed-hearing performance ecosystems.

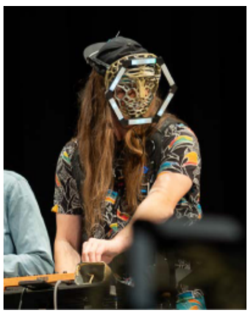
4 Current Practices and Musical Artifacts

Studio sessions allow the ensemble to develop and experiment with custom-designed assistive music technologies while simultaneously integrating commercially available technologies into the studio and performance spaces. The ensemble focused on augmenting these spaces with audio-reactive visuals, audio-haptic sensory integration, and gestural control and expressions. To better conceptualize the findings, in this section, we outline new technologies designed by the team and the existing technologies integrated into the team's multimodal music practices in the studio and on stage.

4.1 Audio Reactive Visuals

Visuals comprise a significant part of DHH individuals' communication means. MB has been developing visual performance systems both to facilitate communication between co-performers, regardless of their hearing, and to visualize the ambient sound. The former was developed during the autoethnography and designed as an LED display to be integrated into parts of musical instruments (such as the guitar fretboard). By retrieving music information, including note onsets, frequency analysis, and parameter mapping, it visualizes the player's musical notes in real-time. This interface was used in performances during the tour on the guitar and drum machines. The latter was previously designed to translate the ambient sound into color and movement using a type of frequency analysis, Fast Fourier Transforms (FFT). By allowing environmental audio to become visible in space, the system aims to amplify sensory awareness of sound and visuals. Figure 1a presents the two systems setup in a rehearsal session during the tour to inform the co-performers of the musical content of the guitarist's performance. Both MB and co-performers can follow the note progressions through the visualizer on the guitarist's fretboard. In the background of Figure 1a, the musician placed the ambient sound visualizer that is connected to the musician's audio output. This system was also integrated into instruments such as pianos and keyboards (Figure 1b), sound mixers and controllers (see Figure 1c), and concert settings (see Figure 1d).

(a)



(b)

(c)

Figure 2: Performers wore audio-reactive masks on the stage during tour concerts, but as a concert artifact and an assistive device for DHH musicians. The image credits belong to Eric Buchner.

DS designed and built a similar LED-based audio-reactive visual assistive technology as a wearable mask, specifically for the ensemble's tour. These 3D printed masks are designed to be worn during live performance, visually mapping music's spectral content across the performer's face. As fully mobile, battery-powered masks, they perform real-time FFT analysis and drive responsive lighting based on the sound input. As a Deaf performer, DS's motivation behind the design derives from two accessibility needs: (1) DS's to read musical qualities, such as rhythm and spectral energy, directly from each other beyond visualizing the sole sonic activity and (2) as an autistic musician, ST's aspiration for performing behind a mask. Figure 2 shows two Deaf musicians, DS and NH, wearing masks during one of the tour performances.

4.2 Audio-Haptic Integration

While visual feedback provides a sensorially rich music information, on-skin vibrotactile haptics provide more direct, nuanced, and articulated musical perception [11, 67]. The ensemble has integrated haptics into their studio and performance practices in multiple ways, two of which have been consistently practiced: (1) tactilely and visually augmented electro-acoustic marimba, EPK [3], and (2) commercially available haptic Woojer vest [4].

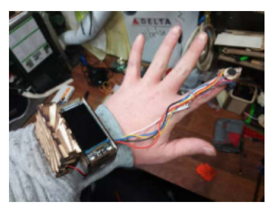
The ensemble leveraged the electro-acoustic marimba's digital sound output by amplifying and streaming the sound signal to the haptic jackets for increased frequency resolution of the pitched percussion. Thanks to this setup, the Deaf musicians can more accurately perceive and follow the music's rhythmic qualities. The haptic jackets were worn by all ensemble members (Figure 4a-4e), regardless of hearing abilities, and by local music collaborators or some of the audience members (Figure 4f-4h). Figure 4 shows that each musician wears the haptic jacket to perceive the same vibrotactile information while on stage or in the studio.

In addition to integrating musical artifacts and assistive technologies, ST customized the studio environment to optimize haptic listening. Adapted from the headphone mixers, the haptic mixers receive the individual sound signals from each instrument without the superfluous sound effects added to the signal. This setup allows the audio engineer and the musicians to customize the audio mix according to the musicians' haptic perception, sonic preferences, and musical needs. The musicians become able to receive and situate each signal directly on the vest and tune their amplification, panning, and mix, providing an increased control and agency.

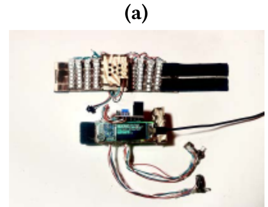
4.3 Gestural Interfaces

As a third modality that supports DHH musicians, gestural interaction in music aligns with sign language expression as a kinesthetic communication method. Led by DS, the ensemble performed with two gestural musical interfaces. DS designed and built a controller system (GeLu) that includes two wearable interfaces: a ring system

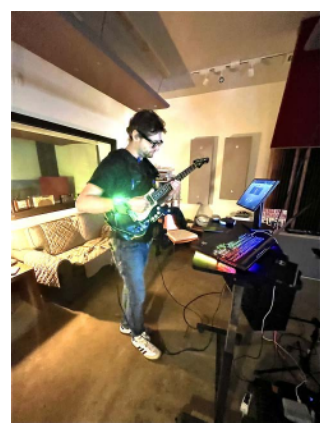
[3]https://www.modemarimba.com/epk/intro
[4]https://www.woojer.com/



(a)

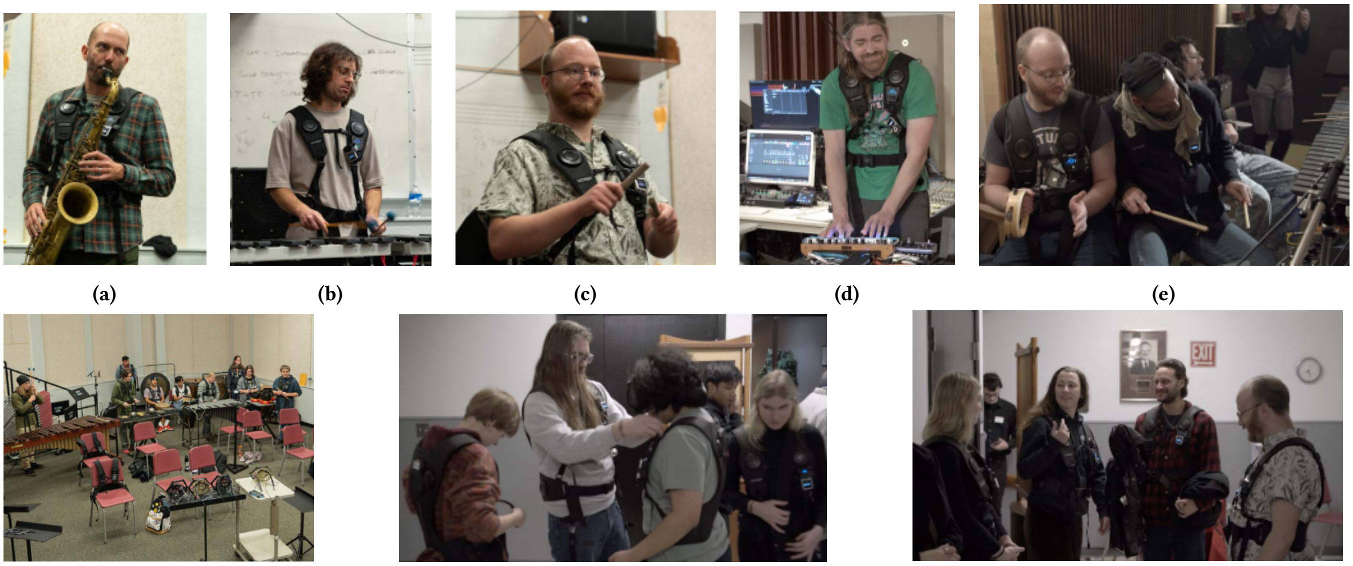


(b)



(c)

Figure 3: A glove-based hand controller for gestural musical interaction aims to accompany American Sign Language (ASL) signs while controlling sound parameters.



(a) **(b)** **(c)** **(d)** **(e)**

(f) **(g)** **(h)**

Figure 4: Haptic vests were worn by the ensemble members both during the studio sessions (4d and 4e) and during the tour concerns (4a-4c). They were also distributed to the audience members. Two Deaf musicians help the audience members wear the haptic suits in 4g and 4h. Image credits belong to Eric Buchner.

with gestural sensing that provides haptic feedback on the fingertips and a bracelet with audio-reactive light feedback (see Figure 3).

These interfaces were originally designed to strengthen the co-performer communication and improvisation between DS and NH, a Deaf music student and CI-user musician to support DHH music students to develop a better understanding of musical rhythm. While one DHH musician performs basic percussive patterns, the other DHH musician receives, through the interface, the sensation of the rhythmic patterns on one fingertip. In return, the musician with the interface can perform polyrhythms using another finger. This performance dynamic mimics the multi-track recording process with headphones that hearing musicians often practice in recording studios, and it instead replaces the audio through headphones with haptics. The design was informed by the signing musicians' need to integrate glove-like technologies into music performance without allowing the interface to disrupt signing.

In addition to facilitating improvisation between two DHH musicians, these interfaces were also used in the studio sessions to inform the development of a DHH music curriculum. However, as early-stage prototypes, they were not performed at the concerts and workshops during the tour. Instead, a more established gestural interface, Theremin, was performed by DS along with other musicians (see Figure 2c).

5 Findings

Through the performance practices over 15 months, we integrated several technologies described in Section 4, which we either adopted and transformed the existing technologies or designed our own assistive technologies and musical artifacts for the personalized needs of DHH musicians. These practices addressed multiple facets of the access barriers, challenges, and opportunities of Deaf musicianship and bespoke design. Some of these aspects overlapped across the arising themes due to the themes' interconnectedness. We organized the findings into the themes of music technology in practice, Deaf musicians' sonic agency and literacy, and the challenges the ensemble faced in mixed-hearing co-creative space, along with the strategies we developed in response to these challenges and limitations.

5.1 Music Technology in *Practice*

LaBelle describes how "sonic agency places an emphasis on the social experience and productions of sound and audition" and how visibility through practice plays a key role in "building alternative frameworks [...] of listening and being heard" [42]. For our practice, participating in music practice and performing on stage provided this visibility, access to listening and audition, and being heard. To be able to improve musicianship skills as DHH musicians and to rehearse in a mixed-hearing performance space, we turn to designing and integrating music technologies into our practices. This practice quickly revealed the most prevalent access need to substitute sound information through vibrotactile haptic feedback, audio-reactive light and visual displays, and gestural interaction.

5.1.1 Haptic Displays. We collectively believe that vibrotactile sensory integration, despite the haptic AT's shortcomings, provides better substitution and better *"demonstration of variance in music" "due to the increased awareness of the sound waves in a specific frequency range,"* i.e. through vibrotactile feedback, specific frequencies can be produced which would be inaudible with traditional speakers otherwise, as ST and NH articulate (4/18/2024).

During all performances, including rehearsals and studio sessions, all musicians wear the haptic vests regardless of their hearing to perceive the same tactile information. ST noted the motivation for providing vibrotactile feedback to all musicians, regardless of whether they have hearing loss. "Bringing haptics into the more conventional performance space, while jazz being an entry point for the Deaf musicians, affords the different aspects of agency sonically and socially in parallel." (11/14/2024). Beyond aiding Deaf performers and facilitating co-performer connections, the haptics also supported the audience's listening experiences. We observed that some audience members were affected by feeling the nuances in the music when listening through the haptic vests. DS retrospectively collected his experience in a Deaf school during an outreach event: *"It seems that they felt nothing as accurate as what NH showed them with the vests. The reaction was pretty strong! They said that they could actually feel things, instead of white noise from many musical instruments getting mixed up. Separating the sounds to left and right made a big difference with those haptics! Like they've just gotten a glimpse of an entirely new dimension that they can start exploring now!"* (4/25/2024).

MB reflects that many Deaf individuals *" have a fractured relationship with sound and music. Many Deaf schools don't have a music program. It has been observed that the concepts of tempo and counting time signatures are often abstract and challenging when teaching rhythm and keeping time. Even in my case, it is difficult for me to tell what key a performance may be in, and I cannot be confident I am playing similar tones unless I have a visual or haptic aid."* (4/19/2024). This fractured relationship originates from missing an introduction to experiencing, understanding, and creating music and sound. For example, NH wrote about his experience demonstrating assistive music technologies in a Deaf school:

> Not a lot of Deaf people engage with music, but when I showed them the haptic vest, they immediately knew what they were missing. They loved loud music, but the opportunity to be loud is scarce. The technology gives them the opportunity to be "loud". (12/3/2024)

Growing up with music while their hearing loss progressed, both MB and NH reflected on the influence of early exposure to music and emphasized how a lack of introduction to the listening experiences impacts DHH individuals' participation and appreciation of music.

5.1.2 Light and Visual Displays. Designing visual assistance derived from specific needs to compensate for different musical elements. MB wrote his motivation for integrating light and visuals for (1) supporting harmonic progression, (2) synchronization, and (3) co-performer dynamics:

> I am currently designing an AI/ML-driven system that would allow me to improvise more confidently with musician peers. It would "listen" to note and chord choices and make recommendations as to different musical possibilities that could be played next. It will visualize these suggestions on instrument-specific LED displays (e.g., guitar fretboards) that light up with the possible "harmonious" suggestions that would evolve as the performer decides which notes/chords to adhere to. The same system would also trigger light and haptic devices that subscribe to the music "session" for others to synchronize their efforts and contribute harmoniously, regardless of hearing ability (4/12/2024).

Similarly, DS's wearable mask design aimed to communicate music's spectral content, or energy, an often neglected musical feature in accessible technology design. DS elaborated how light visuals were connected to sound features:

> Each frequency band has an associated color palette and intensity level. The louder a frequency region is, the more active the mask's corresponding LEDs become, so you can visually read both pitch region and amplitude across someone's face.
>
> This was especially useful in performance. Since Deaf musicians often rely on visual cues to stay in sync, the masks gave us a new layer of feedback. For example, when I played the Theremin, my own mask would react more vividly to its output due to microphone proximity. Likewise, NH and MB had their masks tuned to be more reactive to their own instruments or speakers. The masks allowed us to look at each other and see, not just that someone was playing, but what kind of energy they were putting into it. That's something Deaf performers are often left out of, and the masks helped close that gap (4/14/2024).

These experiences prove the significance of visual perception in DHH individuals', specifically in musicians', everyday experiences. DC noted how much the visual expression and culture were embedded into their communication between hearing and DHH collaborators: *Apart from integrating new technologies into our performance and design spaces, as a hearing person closely collaborating with DHH artists, I have noticed visuals and emotional expression have been big parts of our correspondences. I have received visual references along with my collaborators' experiences, such as videos, visual vernacular, talks, images, or emojis.* (6/17/2024).

5.1.3 Gestural Interfaces. Gestural expressions act as a bridge between embodied aspects of music performance to sign- and visual vernacular-based DHH communication. DS experienced performing with different gestural interfaces during this autoethnography, from intangible instruments such as the Theremin to percussion instruments and custom gestural interfaces. He noted these performance experiences with a strong emphasis on his personal design considerations:

> While I received positive feedback about my control of the Theremin, I personally found it limiting compared to more advanced gestural systems, particularly those I've been prototyping using computer vision or approaches like the Somi [5].
>
> The biggest contrast between the Theremin and GeLu lies in intentionality and feedback. GeLu is built around a feedback loop: haptics, light, motion. That

[5] https://instrumentsofthings.com/

> gives the performer actionable information in real-time. The Theremin, by contrast, offers no tactile feedback and forces the performer into a constrained pose, arms raised, hands shaped unnaturally. As someone who tends to move my entire body while presenting or performing, I pace, sign, and gesture broadly, and I found the Theremin physically stifling (2/7/2025).

He added *"GeLu was intentionally designed to avoid these kinds of limitations. One of my goals has always been to develop gestural interfaces that do not interfere with signing or typing, so performers, engineers, and artists alike can remain fully expressive and functional."* (2/7/2025). DS's reflections highlight that DHH designers value gestural communication and expression, while they struggle to achieve the same expressive affordances with the existing devices. They turn to bespoke designs to meet these needs in desired expressivity and creativity.

5.2 Sonic Agency and Accessibility

MB describes sonic agency as "the ability to influence and interact with sound in a meaningful way, manifesting itself in forms of expression, visual, and tactile experiences as well as creating inclusive environments where all musicians can engage with sound and music. This sense of Deaf-led music agency includes writing music with only vibrations and no audio, incorporating visual tools used also for hearing collaborators, and shifting away from music that relies on listening with the ears towards listening with the body."

This concept relates to sensory inclusivity, extending beyond mere auditory perception and advocating for increased access to music and the sonic world for all individuals, regardless of auditory abilities [42]. NH emphasized the importance of sonic agency and highlighted that DHH individuals have limited access to ''*full understanding of sonic agency in their lives with tools within reach for full equitable experience.*" Similar to NH's remarks, an overarching theme is centered on the separation between DHH individuals' sonic agency and access to resources.

DS articulated this interdependent space and their understanding of sonic agency in their reflections: *"I see sonic agency as the right and the ability to shape sound, whether or not you can hear it. It's about having access to the tools and spaces where sound is created, shared, and experienced. That includes music, but also communication, feedback systems, and control over your environment."* (3/13/2025).

Through group discussion, aligning our experiences, and critically reviewing our observations, we derived three areas where sonic agency is practiced within technology-mediated music performance.

5.2.1 DHH as Music Technology Developers. Because the design space for assistive music technology still lacks the participation of DHH musicians and audio engineers, the existing technologies remain short in their applicability or are unknown to the DHH communities [58]. This challenge encouraged DHH musicians to develop their custom-designed technologies.

Both MB and DS led the team's design practice by reflecting on their lived experiences [34]. MB wrote: *"I have traversed a variety of technologies from analog amplifiers/effect pedals to more recent use of digital amplifier modeling and digital effects. My current process is to use an Axe-FX III to create complex patches of effects that can be tuned to resonate more on the haptic level."* DS's design experience was gained before he started engaging in music practice:

> *I've been a full-stack developer for a decade. I also happened to pick up a lot of embedded hardware and software skills on the side since technology is also my hobby.*

With participation in music and performance, DS's approaches shifted toward more specific considerations:

> As an engineer, I need to be able to type, program, solder, and debug without having to remove or work around a device. That's why I've increasingly moved toward computer vision and contactless systems. This shift to contactless, open-access tools is core to my design philosophy, accessibility that doesn't require sacrificing freedom or complexity (2/7/2025).

While DHH communities face the challenge of resourcing appropriate technologies, embodying the role of designers and technology developers provides an increased sense of sonic agency through gaining "access to the tools and spaces where sound is created, shared, and experienced," as DS emphasized.

5.2.2 Hacking and Re-building as Resilience. Recording studio environments are optimized for normative listening, dampening most of the acoustic vibrations. However, Deaf musicians and producers rely on this knowledge to listen with the whole body. ST and NH realized how NH was utilizing *"the car speakers to give the tactile experience of a concert, but that's what I'm missing when listening to music via streaming or when I am in the studio."* (10/9/2024). For an inclusive studio space, ST repurposed and appropriated existing technologies: *"As other commercially available technologies are getting involved in the practice, the studio setup becomes a blend of customized technology for the specific needs, to make the space more haptically engaging for the deaf musician."* (1/3/2024).

5.2.3 Performance as a Tool. We note that the music for Deaf musicians extends beyond the sole auditory experience and is rather a co-creation of *"a shared musical language that wasn't reliant on hearing"* (MB, 2/27/2025). While for NH, performing with a diverse group of musicians facilitates *"connections across co-performers, motivates, and energizes for future practices"* (2/20/2025), for other DHH performers, performance emphasizes presence and visibility. DS described his experience after the tour:

> For me, performing hasn't made music more meaningful in the emotional sense, but it has sharpened my understanding of how sound is used socially. Music gets attention. [...] It opens doors that other forms of accessibility work just don't. So I've learned to use it as a tool. Not because I care about harmony or melody, but because I care about infrastructure, equity, and building systems that include everyone.

He further articulated a more political aspect of performance to DHH musicians' sonic agency: *"When I talk about sonic agency, I mean the right to be heard, even without making a sound. The right to shape attention. That's what performance gives us, whether through music, sign, lights, or high-tech masks. It's not about sound. It's about presence."*

5.3 Challenges and Strategies in Deaf Music Co-creation

While musicianship is integral in our practices, we recognize that sonic agency extends beyond sound and music-making. It relates to space, tools, timing, movement, and kinesthetic expression, as well as a shared meaning and world-making. This perspective informs how we approach challenges that emerged in mixed hearing co-creative design space, learning and communicating musical concepts, and representing Deaf perspectives in accessibility design research.

5.3.1 Mixed Hearing Performance Ecosystem. Our approach to technology derives from our access needs in music performance but aims to extend to all musicians and listeners, regardless of hearing abilities. While we value the diversity of abilities in the ensemble, we face the challenges of co-creating in a mixed hearing performance ecosystem.

Challenge: Misalignment of Expectations. As NH and DS articulated that there were moments during the tour, the performances *"felt less like a mixed-ability collaboration and more like a hearing music ensemble with a few Deaf performers tacked on"* (3/7/2025). MB echoed this experience of misaligned musical expectations that *"the piece lacked Deaf-centered musicality or structure"* (2/20/2025).

Strategy: Prioritizing Deaf-Composed Music. Sonic agency stems from advocating for Deaf co-creation rather than sole participation as agents to redefine musical standards, language, and appreciation. DS details *"mixed-ability spaces inherently demand strong, Deaf-led coordination and clear cultural grounding. Without that, they default toward hearing norms, and the supposed 'mix' becomes performative rather than collaborative."* (3/27/2025). In the following rehearsal sessions, one of the strategies to better align expectations across different abilities focused on prioritizing deaf-composed musical pieces, regardless of the composer's music level, weaving co-creation into practice and learning.

5.3.2 Learning and Teaching Musical Concepts. Rehearsal sessions in the studios served as an experimental space to practice both learning musical concepts and theory as a Deaf student and teaching music as a hearing or Deaf mentor. Both experiences revealed missed opportunities and challenges with the balance between technology, pacing, and ways of conceptualizing.

Challenge: Unfamiliarity with Musical Concepts. Despite the integration and assistance of technology, DHH musicians tend to experience unfamiliarity with musical concepts and perceptions. Rhythm, as a fundamental musical element, became a challenging concept to perceive and follow with other musicians.

Strategy: Return to the Body. MB composed a musical piece with a rhythmic line resembling heartbeats, "ba-bum, pause, ba-bum" to *"help our Deaf collaborators internalize timing and groove. This somatic approach to rhythm—starting with something you can feel rather than hear—allowed everyone to embody the pulse before we layered additional percussive or melodic elements on top. It offered a grounding rhythm that could be memorized physically, like muscle memory, and became a shared temporal anchor across sensory modalities."* (2/20/2025). Another embodied approach that is adopted by DS during practice sessions addresses the somatic understanding of music through the mimicry of other musicians' musical movements.

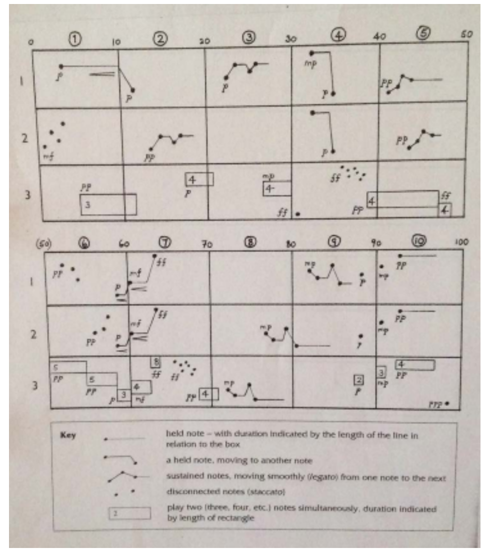


Figure 5: The team adapted the graphic score for their piece "Sounds and Shapes," which was originally composed by Sancton Wood Creative.

Strategy: Graphic Scores as an Accessible Documentation and Notation. NH's compositional explorations in the studio were notated and documented through graphical scores. This approach of communicating musical concepts offered a more accessible transfer of musical knowledge to the Deaf musicians. DS shared, *"I felt more engaged following a graphic 'Sounds and Shapes' score, represented with dots, lines, cells, and rows. I used two different bells, switching to the other for each dot as I led the group."* The graphic score shown in Figure 5 is adapted from Sancton Wood Creative's composition, *Aleatory* [6]. NH worked with his own graphic scores, which were lost during the tour.

5.3.3 Representation in Music Accessibility Design. The main motivation to design custom music technologies is derived from the disconnection between the instruments and technologies that Deaf musicians need and what is publicly available. The existing tool did not always represent Deaf musicians and performers. This lack of representation of Deaf musicianship in design encouraged the team to explore custom and bespoke design as well as appropriating existing technologies.

Challenge: Lack of DHH Music Technologists and Technological Limitations. Despite the increased access and agency that the assistive technologies provide in the performance spaces, the limitations of technologies pose challenges to the ensemble. ST expressed the time lost to technical setups and errors, expressing how

[6]https://sanctonwoodcreative.wordpress.com/2016/10/14/take-a-chance-aleatory-music/

"having all tech available and usable would overcome specific musical challenges such as tones and layering" (3/13/2025). Similarly, NH wrote about the limitations of the existing haptic technologies in providing spatial information about music through haptics in the studio.

Strategy: Bespoke Design and Appropriating Technology. DS and MB, as DHH designers, followed a bespoke approach, integrating their lived experience with technology while still aiming for its implications for all hearing abilities. When technical limitations were faced, the existing technologies were adapted in new ways to address the gap between resources and musical needs. For example, NH and ST created a "haptic design station" in the recording studio using a "traditional headphone mixer" to enable each instrument signal to connect to the performer's preferred listening channel. NH used this new system to map different instruments to the sides of the haptic vest to create musical spatiality on the body.

6 Discussion

Drawing from collaborative group reflections on our lived experiences as a mixed hearing music performance and design team, we outline design implications for different actors involved in musical practice for a more inclusive performance ecosystem.

6.1 Design Implications for Mixed-Hearing Performance Spaces

The majority of performance spaces are occupied by hearing-centric cultures and music communities, creating tensions or misalignment between hearing and DHH musicians. Collaborations and co-creativity in mixed hearing environments require forming new musical worlds collectively [6]. We found embodied, soma-centric approaches center Deaf experience and expand the boundaries of hearing-centric listening practices.

6.1.1 Somatic Approaches to Music Practice. Soma-centric design and somaesthetics [35] prioritize listening to the body and body sensations. Similarly, a somatic approach to music practice shifts the listening from an auditory experience to a bodily listening experience. While such somatic methods become more challenging due to the close integration of senses, they allow musicians to internalize and embody musical concepts [18]. We found the rhythm to suggest multiple methods for an embodied learning approach led by both hearing and Deaf musicians to support Deaf co-performers. Deaf performers often communicate their artistic expressivity through sign language and visual vernacular. Similar approaches, either through sign language incorporation into music performance or gestural interfaces, can support adding physical interaction and embodied layers into performance.

A Deaf performer with little to no musical knowledge developed mimicry as a performance technique to calculate musical timing and improve synchronization. We suggest new compositional strategies to center Deaf music as its own genre and the physicality of the Deaf performance. For example, another Deaf musician composed new music, specifically to amplify the rhythmic elements in music in strong connections to the body's own rhythms, such as heartbeats. Both of these examples highlight the entangled relationship between music perception, Deaf listening, and bodily knowledge. Just as MB emphasizes, music can be internalized, similar to muscle memory.

6.1.2 Deaf Music as a Genre. A possible tension point between hearing and Deaf performance spaces derives from misaligned musical expectations of co-performers. While different subcultures lead their own Deaf-owned music genres, such as Deaf Rave and Deaf hip hop [6, 41, 59], a new form of music needs to be defined by the co-performers. We suggest that performers in mixed hearing collaborations practice attunement exercises to each other's listening modes, technologies, and musical expectations. Such attunement will naturally shift the music away from traditional hearing norms.

6.1.3 Music as a Bridge Across Communities. Given the isolation that Deaf people might experience in a hearing culture, the connections between the two communities are constructed through redefining the music culture, its technologies, and methods to partake. When co-created, music can hold the potential for being a bridge, not across sensory divides but across values and priorities. We suggest collectively engaging in musicking as a shared activity [18]. Our practice-based approach provided the experimental foundations to understand the underlying associations between the body and listening, between movement and music.

6.2 Design Implications for Hearing Assistive Music Technology Design

6.2.1 Promoting for Agency. Trends in designing assistive music technologies follow hearing-led design practices of technologies for the deaf (TFD) [3], even though in music, the need for a Deaf musician's participation emerges more frequently. Music researchers adopted bespoke, participatory, or co-design practices to increase the participation of deaf people in the design process [11, 21, 46]. However, Deaf collaborators' sonic agency remained limited. While hearing designers will need to actively and closely collaborate with deaf designers, a Deaf-led design team would further contribute to increased sonic agency.

6.2.2 Return to the Body. A turn to body-centric approaches in the design practice, specifically for musical expression, will support the shift needed in listening modes from auditory forms to bodily experiences. We need to emphasize that this focus on the body does not need to follow a technology-led trend, but a balance of integrating technology and music theory to enrich the bodily experiences.

6.2.3 Designing within the Deaf Cultures. Designing with Deaf musicians will carry their associated Deaf cultures into the design practice. A hearing designer's attunement to such a culture becomes necessary and inevitable. We find that preserving Deaf communication methods and expressions supports a more inclusive design. For example, drawing parallels between sign language and music provides design considerations that are closely linked with sign language, that both unfold in time, both rely on rhythm, repetition, and silence, and both are performative [13].

Reiterating the importance of bodily experiences, we suggest considerations for what Deaf performers sense. Sensory awareness constitutes a large portion of listening to engage multiple senses in conversation. We recommend anchoring across sensory modalities that Deaf individuals identify, relate with, and appreciate, both sonically and physically.

6.3 Design Implications for Inclusive Recording Studios

Music recording studios are optimized for hearing audio engineers and producers. They fail to support Deaf musicians' existing practices and meet their sensory needs.

6.3.1 Multimodal Assistance. Multisensory integration can multiply the sonic information received by DHH musicians. We find that visual and haptic sensory substitution can both support each modality and amplify the missing sound information while multiplying the quantity of musical information by communicating distinct musical features. The multisensory approach to providing assistive feedback will also support musicians with sensitivities to one modality over others.

6.3.2 Personalizing Haptic Stations. Hearing musicians receive mixing opportunities for individual signals or instruments during performances. In our rehearsals, we replaced this mixing option for musicians' headphones with their haptic listening. This substitution provided flexibility to personalize their haptic feedback on the wearables, allowing for amplification of an instrument or panning multiple instruments on the body. Similar adaptability was provided for the audio-reactive light system to tune into the musicians' own instruments.

6.3.3 Multichannel Communications. In mixed hearing performance settings, in studio or performance stages, the sign language interpreter might not always be visible to all signing musicians. Likewise, verbal instructions or non-verbal co-performer collaborations based on musical cues will create weakened communication in collaborations. Technologies such as closed captions or integrated visuals for each performer could fully support informing musicians. For example, a minimeter was integrated into a Deaf musician's assistive technology setup to inform him of synchronization between co-performers, supporting his musical timing. Such timing cues are often communicated through auditory cues among hearing musicians.

Musical experience and knowledge of a Deaf musician highly depend on musical exposure, love of music, and prior musicality. This interdependence will lead to misinterpretations and misalignment of musical vocabulary (such as rhythm, accent, harmony, or melody) and shared descriptions of sound properties (such as frequency or spectrum). Novice musicians often communicate musical ideas with more descriptive and metaphorical language [58]. An adaptive vocabulary and agreed-upon norms and languages for in-studio collaboration could benefit the co-performer dynamics and their collective experience.

We echo advocating for designing the studio and performance spaces not only with DHH musicians but by DHH musicians, following the shift in accessibility research emphasized by Lewis [43]. We extend Lewis's advocacy with *designing by DHH communities for DHH communities.*

6.4 Limitations and Future Work

We selected autoethnography as a qualitative methodology due to the positionality of each team member and the incorporation of first-person approaches within HCI and music accessibility. The team's personal experiences through ongoing music practice with technology provided opportunities for capturing the personalized, longitudinal, first-person perspective within a practice led by Deaf members. At the same time, the diverse range of abilities and plurality in music and design activities posed limitations on consistency across authors' experiences. Group discussions allowed us to align and repave the research direction collectively as a team.

Authors only shared what they felt comfortable with with the rest of the team to protect the sensitive and personal nature of the autoethnographic data. However, this sensitive process required longer periods of data collection to achieve transparency and honesty [38]. Longitudinal study offers promising results and opportunities to identify larger-scale, overarching patterns while facing changing circumstances in the authors' personal lives. For example, one author's data collection was interrupted due to medical leave, and another one decreased in frequency due to changing schedules and workloads.

Another limitation of the study stems from narrowing the participants down into the core team without including other actors involved in the process, including but not limited to the two other hearing musicians, workshop participants during the tour's outreach events, and CymaSpace's volunteers. Although including the hearing musicians' experiences would provide new perspectives into the phenomenon, we decided that this change would cause a drift in our longitudinal approach and highlight the Deaf voices and experiences.

Lastly, we had to omit large efforts of the team in developing educational materials for Deaf music students to stay within the scope of our research questions. In this research, we framed the strategies developed to support Deaf members of the team. In the future, we plan to incorporate the progress of these educational strategies and their applications at Deaf schools.

7 Conclusion

Through a 15-month-long group autoethnography, we present a case study of a Deaf-owned music and culture institution in a mixed-hearing performance practice. We report our current practices in integrating existing and custom-designed assistive musical technologies into performance practices. We encapsulated nuanced challenges and strategies, self-motivations for new design directions, and tensions and opportunities of co-creating in a mixed hearing performance ecosystem. We present three essential strategies that support DHH individuals' sonic agency and access with or without prior music knowledge. Our case presents how group autoethnography can be applied effectively in a creative, practice-based research setting. We finally provide design suggestions for different actors in the DHH or mixed hearing music performance ecosystem, advocating for diversity in culture, abilities, and methods to be shared among all people.

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